

ANNEXURE 1

Project Gantt Chart

Detailed 11-week timeline (Week 04 – Week 14) — design, implementation & testing phases.

Project	SmartFire Rover — autonomous indoor fire-fighting robot
Team	Md Mohaiminul Hoq Alvi (23101158), Sultan Ruhul Kuddus Nayan (23101159), Ejaj Ahmmod Lemon (23101160)
Course	CSE 316 — Microprocessors & Microcontrollers Lab
Duration	11 weeks (Week 04 – Week 14)

Phase	Task	Owner	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Planning	Problem analysis & literature survey	Team											
Planning	Requirement spec & component selection	Team											
Design	System architecture & block diagram	Nayan											
Design	Circuit design (Tinkercad simulation)	Nayan											
Design	Mechanical / chassis layout	Alvi											
Procurement	Component procurement	Lemon											
Implementation	Hardware assembly & wiring	Alvi											
Implementation	Sensor calibration (IR + DHT11)	Nayan											
Implementation	Firmware: detection + motor control	Nayan											
Implementation	ESP8266 SMTP + GPS integration	Lemon											
Implementation	Servo + pump suppression module	Alvi											
Testing	Unit tests (sensors, comms, drive)	All											

Phase	Task	Owner	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Testing	Integration & field testing	All											
Testing	Performance benchmarking	Lemon											
Documentation	Report writing	Lemon											
Documentation	Presentation & annexure prep	Team											
Documentation	Final demo & defense	Team											

PHASE LEGEND

- Planning
- Design
- Procurement
- Implementation
- Testing
- Documentation

Key Milestones

W5	Design freeze — schematic + BOM finalized
W7	Hardware assembly complete (chassis + wiring)
W10	Firmware end-to-end on prototype
W12	Field-test data collected; benchmark vs baseline
W14	Final report, slides, annexure submitted; demo defended